

The meaning of molecules: A theory of chemical structure

Abstract: In this paper, we defend a formal theory of chemical structure, organized into four components. First, we define an alphabet $\mathcal{A}$ consisting of the 118 known chemical element types, grounded in and characterized by quantum mechanics. Second, we develop a syntax $(\mathcal{A}, \mathcal{K}, \mathcal{P})$, comprising the set of all molecular assemblies $\mathcal{A}$, the set of allowed contexts $\mathcal{K}$, and the set of reaction rules $\mathcal{P}$. Our deep syntax further distinguishes between three levels of chemical equivalence: $\mathcal{M}$ - or $\mathcal{A}$ -equality (at the level of counting atoms and molecules over $\mathcal{A}$); G -isomorphism (at the level of bonding graphs), and ξ -isomorphism (at the level of geometry). Third, we introduce a semantics that assigns meaning to syntactic structures when evaluated under the external thermodynamic parameters $\mathcal{F}$ and $\dot{S}_{\text{prod}}$ and the internal performance parameter B . Fourth, we define an interpretation $\mathcal{I}$ that maps each system to one of at least four thermodynamic regimes: stable, metastable, dissipative, and runaway.

Keywords: Metastable; stable; dissipative; runaway; thermodynamics

§1 Introduction

In this paper, we intend to defend a theory of chemical structure in terms of an alphabet $\mathcal{A}$ or set of 118 chemical element types that are currently known (§ 2), a syntax $(\mathcal{A}, \mathcal{K}, \mathcal{P})$ comprising the set of all assemblies of molecules $\mathcal{A}$, the set of allowed contexts $\mathcal{K}$, and the set of reaction rules $\mathcal{P}$ (§ 2), a semantics that allows us to give meaning to our syntax $(\mathcal{A}, \mathcal{K}, \mathcal{P})$, when a system is evaluated under thermodynamic parameters $\mathcal{F}$ and $\dot{S}_{\text{prod}}$ and an internal performance parameter B (§ 3), and an interpretation $\mathcal{I}$ that allows us to map each system to one of four thermodynamic regimes (stable, metastable, dissipative, runaway) (§ 3).

There are surprisingly few precedents for using the periodic table as the base alphabet for a formal theory of chemical structure. More often than not, when formalisms have been developed, graphs of molecules are taken to be primitive, rather than the base alphabet $\mathcal{A}$ of the 118 known chemical element types (Kerber et al., 2013; Ugi, Stein, and Gruber, 1994). Explicitly defining the 118 elements as the formal alphabet $\mathcal{A}$ from which all chemical syntax is generated therefore appears to be a genuinely distinct starting point. One major advantage of our use of $\mathcal{A}$ as the base alphabet is that it is a finite alphabet that has already been defined and characterized by quantum mechanics (Cao et al., 2019; Schwerdtfeger, Smits, and Pyykkö, 2020). Another major advantage is that our theory gains in elegance without sacrificing rigour, especially since our deep syntax (Def. 4) allows us to distinguish between three levels of equivalence: $\mathcal{M}$ - or $\mathcal{A}$ -equality (at the level of counting atoms and molecules over $\mathcal{A}$); G -isomorphism (at the level of bonding graphs), and ξ -isomorphism (at the level of geometry).

Our syntax $(\mathcal{A}, \mathcal{K}, \mathcal{P})$ of assemblies, contexts, and reaction rules can be

situated within a rich tradition that includes chemical reaction network theory (where chemical systems are characterized in terms of species, complexes, and reaction graphs), chemical organization theory (where chemical organization is defined as a closed and self-maintaining set of components), and assembly theory (where the complexity of molecules and assemblies can be quantified indexically by evaluating the minimal number of steps required to assemble them from fundamental building blocks) (Dittrich and Di Fenizio, 2007; Feinberg, 2019; Marshall, Murray, and Cronin, 2017). Our explicit identification of context $\mathcal{K}$ as a formal component of chemical syntax – distinct from the assemblies and reaction rules – appears to be unique in the context of this tradition.

Our semantics, involving the use of the thermodynamic parameters $\mathcal{F}$ and $\dot{S}_{\text{prod}}$ and an internal performance parameter B , is rooted in Prigogine’s work on dissipative structures and the broader research on non-equilibrium thermodynamics as applied to chemical systems (Prigogine, 1980; Prigogine and Nicolis, 1985; Prigogine and Stengers, 2018). Our claim to theoretical novelty here consists of our separating the syntax (what structures and reactions are possible) and the semantics (what meaning these structures and reactions acquire when evaluated under specific thermodynamic conditions) and our derivation of the internal performance parameter B from the external thermodynamic parameters $\mathcal{F}$ and $\dot{S}_{\text{prod}}$. Last but not least, while thermodynamic regimes are individually well-studied, interest tends to be confined to stable, metastable, and dissipative regimes (Demirel, 2005; Finck, 1947; Mattia and Otto, 2015). The concept of a runaway state – combining strong amplification ($B(\alpha) \gg 1$), high energy flux ($\mathcal{F} \gtrsim \mathcal{F}_{\text{high}}$), and extremely high entropy production ($\dot{S}_{\text{prod}} \gg \dot{S}_{\text{high}}$) – tends to appear more in chemical engineering (thermal runaway) and astrophysics (thermonuclear runaway), and its inclusion in our taxonomy of thermodynamic regimes, framed as an

interpretation layer $\mathcal{I}$ atop the syntax and semantics, is another novel contribution.

Together, the four components – alphabet, syntax, semantics, and interpretation – constitute a unified formal theory of chemical structure that, to the best of our knowledge, is the first to treat the periodic table as a generative base alphabet, while grounding chemical meaning in thermodynamics. For the rest of this paper, we shall develop each component in detail, before showing how these components combine into a coherent and consistent whole.

§2 Alphabet and syntax

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	H 1.008																	He 4.003
2	Li 6.941	Be 9.012																
3	Na 22.99	Mg 24.31											B 10.81	C 12.01	N 14.01	O 16.00	F 18.99	Ne 20.18
4	K 39.10	Ca 40.08	Sc 44.96	Ti 47.88	V 50.94	Cr 52.00	Mn 54.94	Fe 55.85	Co 58.93	Ni 58.69	Cu 63.55	Zn 65.38	Ga 69.72	Ge 72.64	As 74.92	Se 78.96	Br 79.90	Kr 83.80
5	Rb 85.47	Sr 87.62	Y 88.91	Zr 91.22	Nb 92.91	Mo 95.94	Tc 98.91	Ru 101.07	Rh 102.91	Pd 106.42	Ag 107.87	Cd 112.41	In 114.82	Sn 118.71	Sb 121.76	Te 127.60	I 126.91	Xe 131.29
6	Cs 132.91	Ba 137.33	La 138.91	Hf 178.49	Ta 180.95	W 183.84	Re 186.21	Os 190.23	Ir 192.22	Pt 195.08	Au 196.97	Hg 200.59	Tl 204.38	Pb 207.2	Bi 208.98	Po [209]	At [210]	Rn [222]
7	Fr [223]	Ra [226]	Ac [227]	Th 232.04	Pa 231.04	U 238.03	Np 237.05	Pu 244.06	Am 243.06	Cm 247.07	Bk 247.07	Cf 251.08	Es 252.08	Lr 260.11	Fl [289]	Mc [289]	Ts [294]	Og [294]
8				58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.05	71 Lu 174.97	
9				90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np 237.05	94 Pu 244.06	95 Am 243.06	96 Cm 247.07	97 Bk 247.07	98 Cf 251.08	99 Es 252.08	100 Fm 257.10	101 Md [288]	102 No [289]	103 Lr [260]	

Figure 1: The periodic table, generated using the Mendelev Python package (Mentel, 2021)

Def. 1 (Alphabet): Let $\mathcal{A} = \{a_1, \dots, a_n\}$ be a finite set of atoms

$\mathcal{A} = \{\text{H, He, Li, Be}, \dots, \text{Mc, Lv, Ts, Og}\}$ denotes the chemical alphabet or set of 118 chemical element types that are currently known (Fig. 1).

Each type $a \in \mathcal{A}$, uniquely identified by $Z_a \in \{1, \dots, 118\} \subset \mathbb{Z}^+$, is characterized by an observable tuple $\mathcal{O}_a$, whose members include the ionization energy, electron affinity, atomic radius, van der Waals coefficient, and electronegativity. For each element $a \in \mathcal{A}$, the observational tuple $\mathcal{O}_a$ is in principle fully determined by the integer Z_a (number of protons) via the N -electron Schrödinger equation: its components have been systematically derived *inter alia* by Bethe and Salpeter (2013), Cowan (2023), Helgaker, Jorgensen, and Olsen (2013), and Parr (1989), and compiled in the NIST Atomic Spectra Database (Kramida et al., 2024).

Def. 2 (Molecules): The set of molecules is the free commutative monoid on $\mathcal{A}$: $\mathcal{M} = (\mathbb{N}_0^{\mathcal{A}}, +, \mathbf{0})$

$\mathcal{A}$ denotes the chemical alphabet or set of atom types, $\mathbb{N}_0$ denotes all the possible ways of assigning a non-negative integer to each atom type, $+$ denotes how molecules can be combined by adding atom counts, and $\mathbf{0}$ denotes the empty molecule. Molecules $m \in \mathcal{M}$ can therefore be interpreted as finite multisets over $\mathcal{A}$. Caffeine has the molecular formula $\text{C}_8\text{H}_{10}\text{N}_4\text{O}_2$. According to our account of molecules as a commutative bag of atoms, the same atom counts yield the same molecule: $\text{C}_8\text{H}_{10}\text{N}_4\text{O}_2 = \text{H}_{10}\text{C}_8\text{N}_4\text{O}_2 = \text{H}_{10}\text{N}_4\text{C}_8\text{O}_2 = \dots$. Furthermore, caffeine is just:

C: 8

H: 10

N: 4

O: 2

$\mathcal{A} \setminus \{\text{C}, \text{H}, \text{N}, \text{O}\}$: 0

where $\{\text{C}, \text{H}, \text{N}, \text{O}\} \in \mathcal{A}$ and $\{8, 10, 4, 2\} \in \mathbb{N}_0$

Def. 3 (Assemblies): The set of assemblies of molecules is the free commutative monoid on $\mathcal{M}$: $\mathcal{A} = (\mathbb{N}_0^{\mathcal{M}}, +, \mathbf{0})$

Assemblies of molecules $\alpha \in \mathcal{A}$ are finite multisets over $\mathcal{M}$. For instance, ethane (C_2H_6) can be represented as an assembly of two methyl radical (CH_3) molecules. However, neither **Def. 2** nor **Def. 3** includes information about bonding, structure, and geometry. Caffeine and any other molecule with the same molecular formula would therefore be identical and we would be unable to distinguish between isomers. All molecules of identical atomic composition (strings composed of identical letters over $\mathcal{A}$) form an isomer class: they can therefore be further distinguished in terms of either different bonding graphs (constitutional isomerism) or different equilibrium geometries (stereoisomerism).

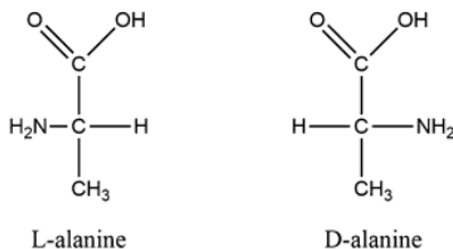


Figure 2: The L- and D-alanine molecules as enantiomers (non-superimposable mirror images) of each other, with different equilibrium geometries (Genchi, 2017, p. 1522)

$\text{C}_2\text{H}_6\text{O}$ is the molecular formula for two different molecules: ethanol and dimethyl ether. These two isomers, however, have different bonding graphs: ethanol’s structural formula ($\text{CH}_3\text{—CH}_2\text{—OH}$) differs from dimethyl ether’s ($\text{CH}_3\text{—O—CH}_3$). In addition, the amino acid molecules L-alanine and D-alanine have exactly the same atomic composition and bonding

graph. Nonetheless, they are distinguished by different equilibrium geometries: the geometric arrangements of their atoms are mirror images of each other (Fig. 2).

Def. 4 (Deep syntax): For a molecule $m \in \mathcal{M}$, a deep syntax on m is the pair $\sigma = (G_\alpha, \xi)$, where:

- (a) $G_\alpha = (V, E, \ell_A, \ell_B)$ is a finite labelled multigraph such that V is a set of atomic sites, E is a multiset of edges (bonds), and ℓ_A and ℓ_B are labelling functions for atoms and bonds respectively;
- (b) ξ assigns spatial coordinates to each atomic site $v \in V$.

Def. 2 consists of pure counting over atoms (composition), while **Def. 3** consists of pure counting over molecules (populations). Their shallow syntax yields free commutative monoids or syntax without relations, making us unable to distinguish between isomers (ethanol versus dimethyl ether, L-alanine versus D-alanine). **Def. 4**, by contrast, provides us with deep syntax: the bonding graph G_α captures constitutional structure (which atoms are bonded and in which order), while the geometry ξ captures spatial structure (how these atoms sit in space, bond lengths, and angles). Two assemblies α_1 and α_2 that have identical composition ($\alpha \in \mathcal{A}$) but distinct bonding graphs (G_α) are constitutional isomers: **Def. 4** allows us to distinguish between ethanol and dimethyl ether. Two assemblies α_1 and α_2 with identical bonding graphs (G_α) but distinct geometries (ξ) are stereoisomers: again, **Def. 4** allows us to distinguish between L- and D-alanine. The deep syntax offered by **Def. 4** allows us to distinguish between three levels of equivalence:

$$\xi\text{-isomorphism} \subsetneq G\text{-isomorphism} \subsetneq \mathcal{M}\text{- or } \mathcal{A}\text{-equality}$$

Chemistry involves more than counting atoms and molecules (**Def. 2**

and **Def. 3**) over the chemical alphabet (**Def. 1**), bonding, and geometry (**Def. 4**). Chemistry involves reactions:

Def. 5 (Reactions): A reaction $r \in \mathcal{R}$ is a pair $r = (\alpha^-, \alpha^+) \in \mathcal{A} \times \mathcal{A}$, where α^- denotes the reactants (what you start with before a reaction) and α^+ denotes the products (what you end with after a reaction). In addition, the atom conservation condition must be satisfied: $\|\alpha^-\| = \|\alpha^+\|$

For instance, when two hydrogen molecules (2H_2) and one oxygen molecule (O_2) react to form two water molecules ($2\text{H}_2\text{O}$), 2H_2 and O_2 are the reactants (α^-) and $2\text{H}_2\text{O}$ is the product (α^+). Atoms are neither created nor destroyed, merely rearranged in accordance with the atom conservation condition in **Def. 5**.

Def. 6 (Reaction rules): A reaction rule is a partial map $\rho : \mathcal{A} \rightarrow \mathcal{A}$ specified by a context $\mathcal{K} \in \mathcal{A}$ and an image $\rho(\mathcal{K}) \in \mathcal{A}$, such that $\|\mathcal{K}\| = \|\rho(\mathcal{K})\|$. This rule ρ acts on any assembly $\alpha \in \mathcal{A}$ with context $\mathcal{K} \subseteq \alpha$ such that $\rho(\alpha) = \alpha - \mathcal{K} + \rho(\mathcal{K})$

Any reaction rule is a partial map, since it has a specific domain of applicability and does not apply to every assembly: if $\mathcal{K} \not\subseteq \alpha$, then $\rho(\alpha)$ is undefined. Each rule describes what to do only when a specific pattern is found. In addition, each reaction rule is typically a local rewrite rule that replaces a matched context $\mathcal{K}$ inside an assembly α by another assembly $\rho(\mathcal{K})$ of the same size. Consider the reaction rule that tells us that if molecules X and Y are present, then they can be replaced by the complex XY . In this instance, the context $\mathcal{K}$ is the part that must be present (molecules X and Y or $X + Y$) and the image $\rho(\mathcal{K})$ is the complex XY . Where the assembly α in question is $X + Y$, the rule will apply to it:

1. $\mathcal{K} = X + Y$ (context)
2. $\rho(\mathcal{K}) = XY$ (image)
3. $\alpha = X + Y$ (assembly) 4. $\rho(\alpha) = \alpha - \mathcal{K} + \rho(\mathcal{K})$ (the reaction rule functioning as a local rewrite operation)
5. $\therefore \rho(\alpha) = (X + Y) - (X + Y) + XY = XY$ (product; from 1, 2, 3, and 4)

In a similar vein, where $\alpha = Z + X + Y$, the product $\rho(\alpha)$ will be $Z + XY$. By contrast, as there is no match between $\mathcal{K}$ and $\alpha = X + Z$, the reaction rule will not apply in the latter instance. The three primitive reaction rules ($\mathcal{K} \mapsto \rho(\mathcal{K})$) are:

(Synthesis) $A + B \mapsto AB$;

(Decomposition) $AB \mapsto A + B$

(Catalyzed reaction) $A + B + C \mapsto AB + C$, where C is the catalyst

The reachable chemistry from α_0 under a set of reaction rules $\mathcal{P}$ is:

$$\mathcal{S}(\alpha_0, \mathcal{P}) = \{\alpha \in \mathcal{A} \mid \alpha_0 \xrightarrow{\mathcal{P}^*} \alpha\}$$

$\mathcal{S}(\alpha_0, \mathcal{P})$ denotes the set of all assemblies reachable from the initial state α_0 by a finite sequence of rule applications. This also represents the closure of α_0 under the dynamics generated by the rule set $\mathcal{P}$. More generally, where $\mathcal{A}$ denotes the set of all assemblies of molecules, $\mathcal{K}$ denotes the set of allowed contexts (patterns that can be matched), $\mathcal{P}$ denotes the set of reaction rules (allowed transformations on matched patterns), $(\mathcal{A}, \mathcal{K}, \mathcal{P})$ yields the full syntactic data of the system.

§3 Semantics

Def. 1, **Def. 2**, and **Def. 3** respectively yield the alphabet $\mathcal{A}$, set of molecules $\mathcal{M}$, and the set of assemblies of molecules $\mathcal{A}$ (§ 2). **Def. 4**, in turn, provides us with the deep syntax for characterizing any assembly $\alpha \in \mathcal{A}$ in terms of counting atoms and molecules over the chemical alphabet, bonding graphs (G_α), and geometries (ξ) (§ 2). **Def. 5** and **Def. 6** allow us to make sense of reactions and reaction rules in terms of a set of allowed contexts ($\mathcal{K}$), a set of reaction rules ($\mathcal{P}$), and the rules themselves as local rewrite rules that satisfy the atom conservation condition. Together, **Def.s 1–6** support a well-defined triple $(\mathcal{A}, \mathcal{K}, \mathcal{P})$, capturing a grammar $(\mathcal{K}, \mathcal{P})$ on the universe $\mathcal{A}$ of assemblies of molecules.

Semantics gives physical meaning to our formal syntax $(\mathcal{A}, \mathcal{K}, \mathcal{P})$ by evaluating a system under two external thermodynamic parameters:

Def. 7 (Thermodynamic parameters):

Def. 7A (Energy flux): External energy flux $\mathcal{F}$ refers to the rate at which energy enters a system from its surroundings. $\mathcal{F}$ drives reactions and keeps a system away from equilibrium;

Def. 7B (Entropy production rate): Entropy production rate $\dot{S}_{\text{prod}}$ refers to the rate at which entropy (heat or dissipated energy) is produced inside a system due to irreversible processes.

External energy flux $\mathcal{F}$ is non-negative: there is a ban on negative energy entering a system. Likewise, entropy production rate is non-negative: this is consistent with the second law of thermodynamics:

$$(\mathcal{F}, \dot{S}_{\text{prod}}) \in \mathbb{R}^{\geq 0} \times \mathbb{R}^{\geq 0}$$

A third parameter emerges from the first two thermodynamic parameters defined in **Def. 7**:

Def. 8 (Internal rate functions):

Def. 8A (Rate of internal resource production): $P(\alpha)$ denotes the rate at which an assembly α produces internal resources;

Def. 8B (Rate of internal resource consumption): $C(\alpha)$ denotes the rate at which α consumes internal resources.

Def. 8A and **Def. 8B** allow us to infer a ratio measuring the balance of production versus consumption within a system:

$B(\alpha) = \frac{P(\alpha)}{C(\alpha)} \in \mathbb{R}^{\geq 0}$, relative to the allowed contexts or patterns specified under $\mathcal{K}$

If $P(\alpha) > C(\alpha)$, the assembly α is net-productive ($B(\alpha) > 1$). If $P(\alpha) = C(\alpha)$, α is balanced ($B(\alpha) = 1$). If $P(\alpha) < C(\alpha)$, α is net-consuming ($B(\alpha) < 1$). Our formal syntax $(\mathcal{A}, \mathcal{K}, \mathcal{P})$ specifies what transformations are allowed within our universe of assemblies of molecules $\mathcal{A}$, while the external thermodynamic parameters $(\mathcal{F}, \dot{S}_{\text{prod}})$ specify the energetic environment in which these transformations unfold. The external thermodynamic parameters $(\mathcal{F}, \dot{S}_{\text{prod}})$, in turn, allow us to infer an internal performance parameter B , from which we can determine whether an assembly α is net-productive ($B(\alpha) > 1$), balanced ($B(\alpha) = 1$), or net-consuming ($B(\alpha) < 1$). We are now ready to define our semantics:

Def. 9 (Interpretation): An interpretation $\mathcal{I}$ is a map of the form:

$$\mathcal{I} : (\mathcal{A}, \mathcal{K}, \mathcal{P}) \times (\mathcal{F}, \dot{S}_{\text{prod}}, B) \rightarrow \{\text{A, B, C, D}\}$$

This interpretation $\mathcal{S}$ allows us to assign each system under thermodynamic conditions to one of at least four thermodynamic regimes A (stable), B (metastable), C (dissipative), or D (runaway). Where $\mathcal{F}_{\text{low}}$, $\mathcal{F}_{\text{moderate}}$, $\mathcal{F}_{\text{high}}$, $\dot{S}_{\text{low}}$, $\dot{S}_{\text{moderate}}$, and $\dot{S}_{\text{high}}$ denote explicit thermodynamic thresholds:

Regime	Flux	Entropy production rate	Internal performance parameter	Description
Stable (A)	$\mathcal{F} \leq \mathcal{F}_{\text{low}}$	$\dot{S}_{\text{prod}} \leq \dot{S}_{\text{low}}$	$B(\alpha) \approx 1$	Energy minimum (global equilibrium state)
Metastable (B)	$\mathcal{F}_{\text{low}} \lesssim \mathcal{F} \lesssim \mathcal{F}_{\text{moderate}}$	$\dot{S}_{\text{prod}} \leq \dot{S}_{\text{low}}$	$B(\alpha) \gtrsim 1$	Energy barrier (kinetically trapped, local equilibrium state)
Dissipative (C)	$\mathcal{F} \gtrsim \mathcal{F}_{\text{high}}$	$\frac{\dot{S}_{\text{moderate}}}{\dot{S}_{\text{prod}}} \lesssim \frac{\dot{S}_{\text{moderate}}}{\dot{S}_{\text{high}}}$	$B(\alpha) > 1$	Regulated flow (flux-maintained state)
Runaway (D)	$\mathcal{F} \gtrsim \mathcal{F}_{\text{high}}$	$\dot{S}_{\text{prod}} \gg \dot{S}_{\text{high}}$	$B(\alpha) \gg 1$	Runaway state

Table 1: The four thermodynamic regimes A, B, C, and D to which any system under thermodynamic conditions can be assigned

Systems that have been assigned to Regime A are stable structures at thermodynamic equilibrium, corresponding to the global minimum of Gibbs free energy G ($\arg \min G$). As these structures are already at the lowest accessible free energy state, there is nowhere thermodynamically favourable to go. Any perturbation increases G in such a way that, given sufficient time, these structures are returned to the minimum. CO_2 and H_2O are examples of Regime A structures, since their covalent bonding geometry comprises the lowest free energy configurations of C, O and H, O under ambient conditions. NaCl are another example of a Regime A structure, since its crystalline geometry is characterized by electrostatic interactions between Na^+ and Cl^-

ions in the lattice that minimize G .

Metastable structures are excluded from Regime A and assigned to Regime B instead, where they occupy only local minima separated from stable, lower-energy states by an activation energy barrier. Typically, the activation energy needed to disrupt metastable structures vastly exceeds the available thermal energy: $\Delta E^\ddagger \gg k_B T$, where k_B denotes the Boltzmann constant. Regime B structures persist not because they are thermodynamically favoured, but rather because the pathway to a lower-energy state remains kinetically inaccessible. Diamonds are an example of a kinetically trapped, metastable structure: each carbon atom is bonded to four other carbon atoms in a tetrahedral sp^3 network (Fig. 3). Graphite is more thermodynamically stable than diamond: each carbon atom is bonded to three other carbon atoms in a planar sp^2 network stacked in layers (Fig. 3). However, as converting diamond to graphite requires breaking many strong bonds at once, the activation energy barrier ($\Delta E^\ddagger$) is so high that this transformation would take longer than the age of the universe.

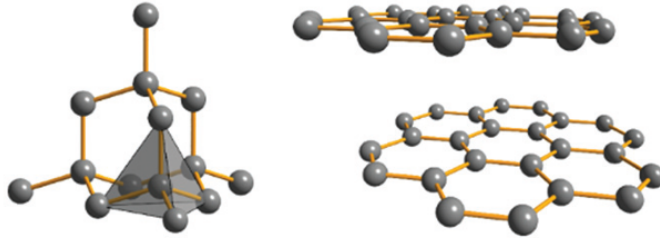


Figure 3: From left to right: The different crystal structures of a diamond ($\mathcal{R}_C$ or tetrahedral sp^3 network) and graphite ($\mathcal{R}_C^*$ or planar sp^2 network) (Popov et al., 2019, p. 10961)

Hydrocarbons, another example of Regime B structures, include methane (CH_4), ethane (C_2H_6), propane (C_3H_8), and benzene (C_6H_6). Hydrocarbons initially comprise C — C and C — H bonds. Combustion can trans-

form these hydrocarbons into thermodynamically favoured structures: the $\text{C}=\text{O}$ and $\text{O}-\text{H}$ covalent bonds in CO_2 and H_2O respectively. However, as activation energy is required to initiate the combustion, hydrocarbons persist in their metastable structures until the energy barrier is overcome.

Regimes A and B are primarily determined by the thermodynamic energy landscape: A corresponds to stable global minima, while B corresponds to metastable local minima. CO_2 , H_2O , NaCl , diamonds, and hydrocarbons are best regarded as equilibrium or near-equilibrium structures selected by the Gibbs free-energy landscape under the relevant thermodynamic conditions. By contrast, Regime C structures are better described as non-equilibrium, steady-state structures whose persistence depends on regulated flows and ongoing fluxes being maintained. Turing (1990) patterns are an example of Regime C structures: a chemical (reaction-diffusion) flux $\mathcal{F} \gtrsim \mathcal{F}_{\text{high}}$ must be sustained to keep these Turing patterns going.¹ After all, these Turing patterns continually dissipate energy into their environment via diffusion and reaction, just as $\mathcal{F}$ replenishes this energy: $\dot{S}_{\text{moderate}} \lesssim \dot{S}_{\text{prod}} \lesssim \dot{S}_{\text{high}}$. If $\mathcal{F}$ is abruptly removed, then these Turing patterns will decay diffusively back to a uniform equilibrium state.

Bénard cells offer another instance of Regime C (dissipative) structures.

¹ Here is the full set-up for Turing patterns: suppose there is a spatially extended chemical system, comprising at least two interacting substances (an activator u and an inhibitor v), and the diffusion coefficients D_u and D_v . Before any chemical flux $\mathcal{F}$ is applied, the concentrations of u and v will be uniform and the chemical system remains homogeneous. As long as the parameters (diffusion coefficients, reaction rates, and so on) remain within a certain regime, small perturbations will decay and the chemical system will return to its homogeneous steady state. However, when the inhibitor diffuses sufficiently faster than the activator ($D_v \gg D_u$), this results in local activation (u reinforces itself) and long-range inhibition (v diffuses outward fast and suppresses the neighbouring regions). The original homogeneous state becomes unstable, the chemical system undergoes diffusion-driven chemical instability, and regular spatial patterns (for instance, stripes, spots, and so on) will emerge, breaking translational symmetry.

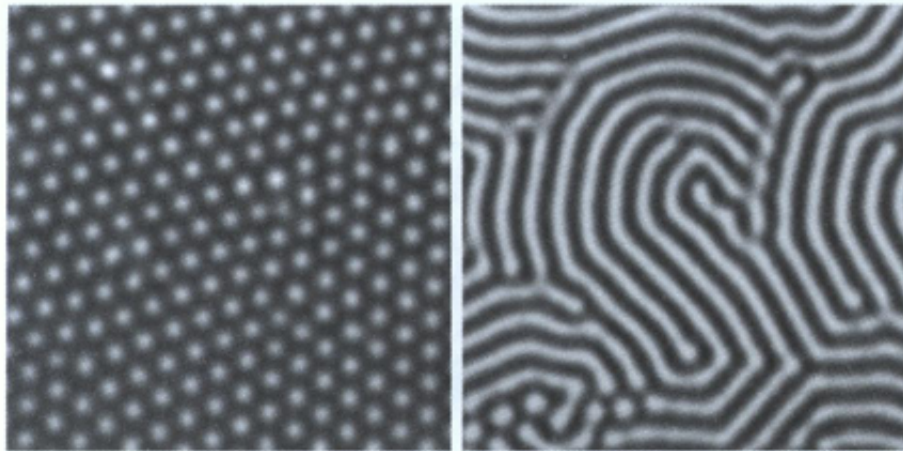


Figure 4: The generic patterns of an activator-inhibitor scheme. Images courtesy of Jacques Boissonade and Patrick De Kepper (University of Bordeaux) and reproduced in Ball (2015, p. 6)

In fluid thermodynamics, Rayleigh–Bénard convection refers to a type of natural convection, occurring in a horizontal layer of fluid that is heated from below, giving rise to a regular hexagonal pattern of convection cells known as Bénard cells.² Once the imposed temperature gradient is sufficiently large ($\Delta T > \Delta T_c$), the system organizes into a hexagonal pattern of convection cells that transport heat more efficiently. This pattern exists only because Bénard cells continuously receive energy from the temperature gradient and continuously dissipate it. These dissipative structures are therefore characterized by sustained flux ($\mathcal{F} \gtrsim \mathcal{F}_{\text{high}}$) and increased entropy production

² Here is the full set-up for Bénard cells: the horizontal layer of fluid can be regarded as a layer between two parallel planes. Before heat is applied from below to the bottom plane, the temperature of the top plane will be the same as that of the bottom plane, and the liquid will be uniform. As the temperature gradient ΔT between the two planes remains below a critical threshold ($\Delta T < \Delta T_c$), the liquid will continue to be uniform, conducting heat purely by diffusion. However, as the temperature gradient exceeds this critical threshold ($\Delta T > \Delta T_c$), the heated fluid will spontaneously organize into a regular hexagonal pattern of convection cells, breaking translational symmetry in the horizontal plane.

($\dot{S}_{\text{moderate}} \lesssim \dot{S}_{\text{prod}} \lesssim \dot{S}_{\text{high}}$). Again, if $\mathcal{F}$ is abruptly removed, then Bénard cells will decay diffusively back to their original equilibrium state.

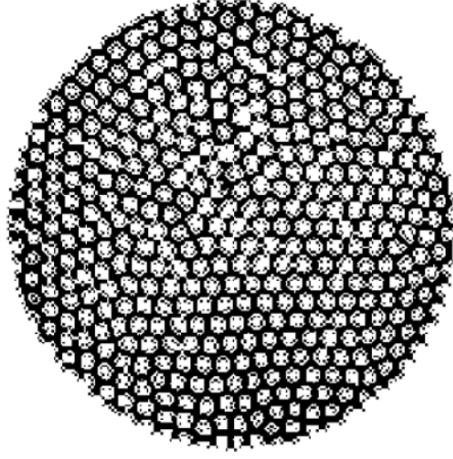


Figure 5: One of Bénard’s celebrated original photographs of the top view of convection patterns in a thin layer of spermaceti, heated from below (Manneville, 2006, p. 42)

Living cells are yet another example of Regime C (dissipative) structures. Nutrient, light, and redox gradients provide the sustained external flux ($\mathcal{F} \gtrsim \mathcal{F}_{\text{high}}$), while metabolic networks convert the energy input into work, heat, dissipated energy, and waste, thereby producing and exporting entropy ($\dot{S}_{\text{moderate}} \lesssim \dot{S}_{\text{prod}} \lesssim \dot{S}_{\text{high}}$). The internal organization of living cells is maintained not by the minimization of Gibbs free energy G . Rather, it is sustained by channelling the dissipation of energy into a flux-driven, steady state that is far from thermodynamic equilibrium. Indeed, the entropy production is sufficiently high to support the structure, but not too high such that the structure gets destroyed. All things considered, Regime C structures such as Turing patterns, Bénard cells, and living cells are high-throughput, entropy-processing systems that maintain their local order dissipatively by exporting entropy to their surroundings.

Whereas Regime C structures maintain their local order through dissipation, Regime D structures are unable to contain this dissipation. Regulatory balance becomes upset, entropy production grows in an uncontrolled manner, and structures become unstable, fragmented, or self-destroying. Flux remains high ($\mathcal{F} \gtrsim \mathcal{F}_{\text{high}}$), while entropy production is extremely high and unregulated ($\dot{S}_{\text{prod}} \gg \dot{S}_{\text{high}}$). An example of a Regime D structure is the decomposition of ammonium nitrate (NH_4NO_3), a widely used fertilizer and explosive material, which can become violent under confinement, contamination, or strong heating. It can transition from a rapid chain reaction to explosion, driven by a self-amplifying reaction network ($B(\alpha) \gg 1$), high energy flux ($\mathcal{F} \gtrsim \mathcal{F}_{\text{high}}$), and extremely high entropy production ($\dot{S}_{\text{prod}} \gg \dot{S}_{\text{high}}$).

Another example of a Regime D structure is a cytokine storm, a severe immune reaction that releases too many cytokines, causing life-threatening inflammation, organ failure, and even death. As cytokines activate immune cells, even more cytokines are released, such that $B(\alpha) \gg 1$. The metabolic activity and inflammatory signalling of the immune system are strongly driven, although the resulting flux is no longer organized into any stable, dissipative pattern. Instead, the inflammatory feedback becomes self-amplifying, unregulated, and life-threatening. Associated with high energy flux ($\mathcal{F} \gtrsim \mathcal{F}_{\text{high}}$), cytokine storms are runaway inflammatory states in which the metabolic activation of immune cells, cytokine signalling, and inflammatory cell death reinforce one another. This results in a self-amplifying, destructive regime, with strong metabolic throughput and a loss of regulatory control.

If $\mathcal{P}$ denotes the set of reaction rules, then the subset $\mathcal{P}_{\text{trans}} \subsetneq \mathcal{P}$ encodes those rules that allow systems to move between regimes:

A $\rightarrow$ B: From a global equilibrium to a local minimum separated

by an activation energy barrier (for instance, water supercooled below 0° C is metastable relative to its lowest Gibbs free energy state, ice);

A $\rightarrow$ C: From a global equilibrium to a regulated dissipative state (for instance, a fluid layer heated from below transitioning into Bénard convection);

A $\rightarrow$ D: From a global equilibrium to runaway instability (for instance, a batch reactor thermal runaway after cooling failure);

B $\rightarrow$ A: From a local minimum into a true global minimum (for instance, a supercooled liquid crystallizing into its stable solid phase);

B $\rightarrow$ C: From a local minimum (kinetically trapped) to a regulated dissipative state (for instance, a cell-like system becoming sustained by continuous energy and matter throughput and transitioning into a flux-driven, dissipative state far from equilibrium);

B $\rightarrow$ D: From a local minimum (kinetically trapped) to runaway instability (for instance, a stored chemical mixture that remains metastable until a perturbation launches the thermal runaway);

C $\rightarrow$ A: From a dissipative state to a global equilibrium (for instance, when the temperature gradient is removed from Bénard cells and the convection pattern dies out);

C $\rightarrow$ B: From a dissipative state to a local minimum (for instance, when tardigrades enter into cryptobiosis: metabolic activity is minimized and biological structure is preserved in a metastable configuration);

C $\rightarrow$ D: From a dissipative state to runaway instability (for instance, when a metabolic network becomes overdriven, loses homeostatic regulation, and enters a pathological regime marked

by self-amplifying dynamics);

$D \rightarrow C$: From a runaway state to a dissipative state (for instance, when feedback control reduces an overdriven biochemical cascade into a non-pathology, flux-maintained steady state);

$D \rightarrow B$: From a runaway state to a local minimum (kinetically trapped) (for instance, when a damaged biological network collapses into a low-activity, metastable state);

$D \rightarrow A$: From a runaway state to a global equilibrium (for instance, when a decomposing system is stabilized and cooled to the global minimum configuration).

§4 Conclusion

In conclusion, we have defended a theory of chemical structure that comprises the following:

An alphabet $\mathcal{A}$ or set of 118 chemical element types that are currently known (**Def. 1**);

A syntax $(\mathcal{A}, \mathcal{K}, \mathcal{P})$ comprising the set of all assemblies of molecules $\mathcal{A}$, the set of allowed contexts $\mathcal{K}$, and the set of reaction rules $\mathcal{P}$ (**Def.s 2–6**);

A semantics that allows us to give meaning to our syntax $(\mathcal{A}, \mathcal{K}, \mathcal{P})$, when a system is evaluated under thermodynamic parameters $\mathcal{F}$ and $\dot{S}_{\text{prod}}$ and an internal performance parameter B (**Def.s 7–9**);

An interpretation $\mathcal{I}$ that allows us to assign each system to one of four thermodynamic regimes (stable, metastable, dissipative, runaway) (Table 1).

The principal theoretical novelties of our framework include the explicit

use of the periodic table as a finite base alphabet $\mathcal{A}$; the formal separation of the syntax, semantics, and interpretation as distinct yet interlocking layers; and the inclusion of the runaway regime as a fourth regime alongside the more familiar stable, metastable, and dissipative regimes. We regard our theory not as a finished product but rather a precise starting point, open to scrutiny and further refinement.

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